

psyc2021

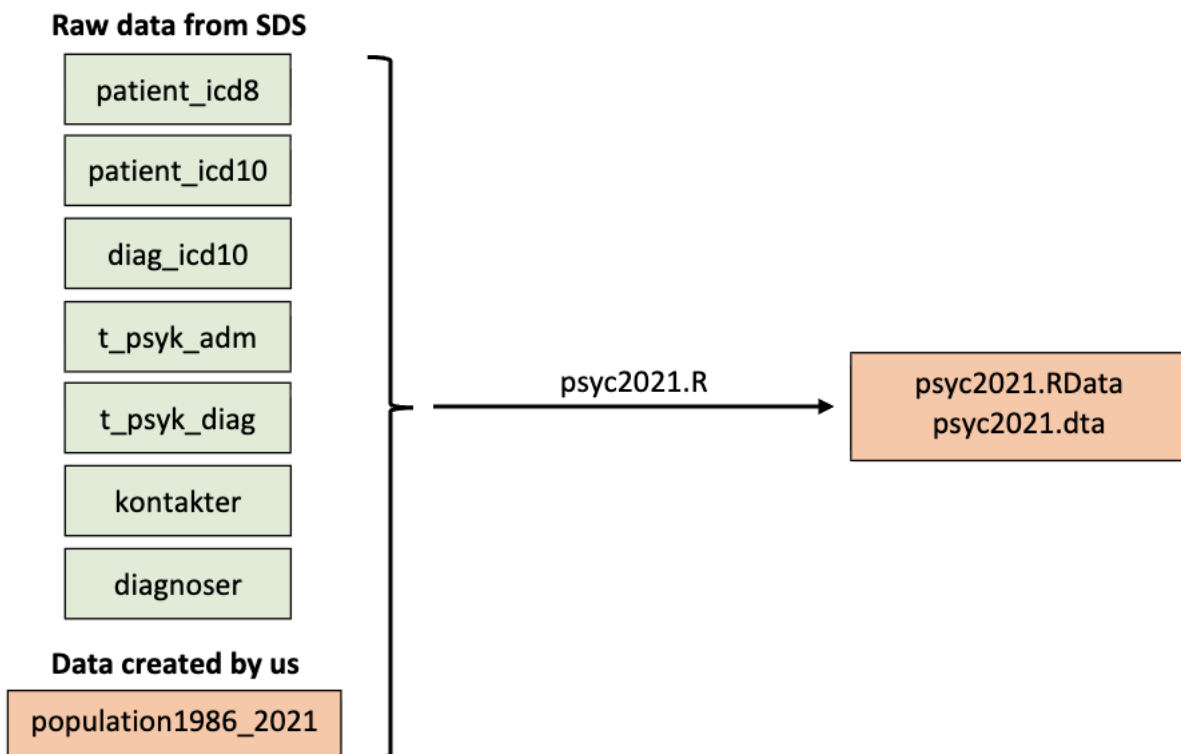
Version: 2 (September 30, 2024)

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Changes from previous version: we fixed a coding mistake that would classify around 0.4% of inpatient contacts as outpatient or emergency room visits. Those misclassified were based on LPR3 (years 2019-2022), and hospital contacts before 2019 were not affected by this mistake.

The dataset `psyc2021` contains information on all diagnoses of mental disorders in Danish psychiatric departments until August 24, 2022 for all individuals living in Denmark between January 1, 1986, and December 31, 2021. There is one observation for each diagnosis, including date of diagnosis, type of patient, and specific diagnosis code.

Code: psyc2021.R (R code)
Output: psyc2021.RData (R file)
 psyc2021.dta (Stata file)



The dataset is created based on the following registers:

- Register `population1986_2021`, which includes basic information on all individuals living in Denmark between January 1, 1986, and December 31, 2021. We have created this dataset based on the registers from [Statistics Denmark \(DST\)](#) on population ([‘Befolkning’; BEF](#)) and deaths in Denmark ([‘Døde i Danmark’; DOD](#)). Specific details and R code are available on OSF (<https://osf.io/zhfyp/>).
- The Danish Psychiatric Central Research Register,¹ which contains information on admissions to psychiatric inpatient facilities since April 1969 and visits to outpatient

psychiatric departments and emergency departments since January 1, 1995. From January 1, 1970, all psychiatric departments were committed to report the data, and it is the first year with valid data. The diagnostic system used is the Danish modification of the *International Classification of Diseases, Eighth Revision* (ICD-8) from 1969 to 1993, and *Tenth Revision* (ICD-10) from 1994 onwards. The register is available through different specific registers from the [Danish Health Data Authority \(Sundhedsdatastyrelsen; SDS\)](#):

- The Danish Psychiatric Central Research Register ([‘Det Psykiatriske Centrale Forskningsregister’; PCR](#)) from 1969 to 1994 (dataset `patient_icd8` from 1969 to 1993; datasets `patient_icd10` and `diag_icd10` in 1994).
- The Danish National Patient Register² version LPR2 ([‘Landspatientregisteret’; LPR2](#)) from 1995 to March 2019 (datasets `t_psyk_adm` and `t_psyk_diag`).
- The Danish National Patient Register² version LPR3 ([‘Landspatientregisteret’; LPR3](#)) from February 2019 onwards (datasets `kontakter` and `diagnoser`). [According to SDS](#), there is a combination of LPR2 and LPR3 data between January and March 2019.

Dataset `psyc2021` contains all diagnoses of mental disorders (ICD-10 codes F00-F99; ICD-8 codes 290-315) classified as main diagnoses (*hoveddiagnoser/aktionsdiagnoser*; type ‘A’), auxiliary diagnoses (*bidiagnoser*; type ‘B’), basic diagnoses (*grundmorbus*; type ‘G’), and associated diagnoses (*tillægsdiagnoser*; type ‘+’). The dataset does not include referral diagnoses (*henvisningsdiagnose*; type ‘H’) or diagnoses linked to a hospital contact in which the main diagnosis is related to healthy individuals (i.e., ICD-8 code Y719; ICD-10 codes Z03.2, Z03.8, Z03.9, Z76.3).

The final dataset contains 12,169,469 observations from 1,056,908 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits’ positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.
- **sex**: Sex based on DST’s variable [koen](#) (1: male; 2: female), indicating the legal sex of the individual based on information from the CPR number. For more information, please look at documentation for register `population1986_2021` (available on OSF; <https://osf.io/zhfyp/>).
- **birth_d**: Date of birth based on DST’s variable [foed_dag](#).
- **date_dx**: Date of diagnosis.
- **date_ending**: Date of discharge.^a
- **type_patient**: Type of patient (1: inpatient; 2: outpatient; 3: emergency room visit).^b
- **length_hours**: Length of the hospital visit in hours (only available for LPR3 to identify the type of patient).^b
- **diag**: ICD code for diagnosis (in ICD-8 or ICD-10).
- **type_diag**: Type of diagnosis (A: main; B: auxiliary; G: basic; +: associated).

- **MDatlas10:** This variable classifies the diagnosis “diag” into 10 broad categories of mental disorders as described in Table 1. Some diagnosis might not be classified into any of the categories.
- **recnum:** Identifier of hospital contacts.
- **lpr:** LPR version (2: LPR2; 3: LPR3).
- **icd:** ICD version (8: ICD-8; 10: ICD-10).

^a According to [the Danish Health Data Authority guidelines](#), date of discharge for outpatients is registered on the final day for a course of treatments, and can occur months after first contact. Date of discharge for inpatients is when they leave hospital. If no date of discharge is registered for emergency room visits, date of discharge should be set to date of first contact (date of diagnosis).

^b While the LPR2 (used until March 2019) has a clear definition of type of patient, this is not the case for LPR3. For LPR3, we followed the guidelines on how to classify patients according to the [Danish Health Data Authority suggestions](#) and a previous publication comparing LPR2 with LPR3 data.³ In brief, patients were classified as ‘inpatients’ if they stayed at the hospital for 8 hours or longer; ‘emergency room visits’ if they were at the hospital for less than 8 hours and they were attended in the emergency unit (*skadestue*) and priority set to acute patients (*ATA1*); or ‘outpatient’ otherwise. The variable ‘length_hours’ can be used to easily change the definition between inpatients and emergency room visits.

Table 1. Classification of mental disorders into 10 broad categories in variable MDatlas10. This classification is slightly adapted from a previous publication using Danish registers.⁴

	Broad categories of mental disorders	ICD-10	ICD-8
D00	Organic, including symptomatic, mental disorders <i>Includes dementia in Alzheimer’s disease, vascular dementia, etc.</i>	F00 – F09	290.09, 290.10, 290.11, 290.18, 290.19, 292.x9, 293.x9, 294.x9 (except 294.39), 309.x9
D10	Mental and behavioral disorders due to psychoactive substance use <i>Includes use of alcohol, cannabis, cocaine, nicotine, opioids, sedatives, hypnotics, anxiolytics, etc.</i>	F10 – F19	291.x9, 294.39, 303.x9, 303.20, 303.28, 303.90, 304.x9
D20	Schizophrenia and related disorders <i>Includes schizophrenia, schizotypal disorders, schizoaffective disorders and other psychotic disorders.</i>	F20 – F29	295.x9, 296.89, 297.x9, 298.29-298.99, 299.04, 299.05, 299.09, 301.83
D30	Mood disorders <i>Includes bipolar disorder, depressive disorders, etc.</i>	F30 – F39	296.x9 (excluding 296.89), 298.09, 298.19, 300.49, 301.19
D41	Neurotic, stress-related, and somatoform disorders <i>Includes anxiety disorders, phobias, obsessive-compulsive disorders, etc.</i>	F40 – F48	300.x9 (excluding 300.49), 305.x9, 305.68, 307.99
D51	Eating disorders <i>Includes anorexia nervosa, bulimia nervosa, etc.</i>	F50	306.50, 306.58, 306.59

D61	Specific personality disorders	F60	301.x9 (excluding 301.19), 301.80, 301.81, 301.82, 301.84
D70	Intellectual disabilities	F70 – F79	311.xx, 312.xx, 313.xx, 314.xx, 315.xx
D81	Pervasive developmental disorders <i>Includes autism spectrum disorder</i>	F84	299.00, 299.01, 299.02, 299.03
D91	Behavioral and emotional disorders with onset usually occurring in childhood and adolescence <i>Includes attention-deficit hyperactivity disorder, conduct disorders, childhood emotional disorders, etc.</i>	F90 – F98	306.x9 (except 306.59), 308.0x

1. Mors O, Perto GP, Mortensen PB. The Danish Psychiatric Central Research Register. *Scand J Public Health*. 2011;39(7 Suppl):54-57. doi:10.1177/1403494810395825
2. Lynge E, Sandegaard JL, Rebolj M. The Danish National Patient Register. *Scandinavian Journal of Public Health*. 2011;39(7 Suppl):30-33. doi:10.1177/1403494811401482
3. Bernstorff M, Hansen L, Perfalk E, Danielsen AA, Østergaard SD. Stability of diagnostic coding of psychiatric outpatient visits across the transition from the second to the third version of the Danish National Patient Registry. *Acta Psychiatrica Scandinavica*. 2022;146(3):272-283. doi:10.1111/acps.13463
4. Pedersen CB, Mors O, Bertelsen A, et al. A comprehensive nationwide study of the incidence rate and lifetime risk for treated mental disorders. *JAMA Psychiatry*. 2014;71(5):573-581. doi:10.1001/jamapsychiatry.2014.16

